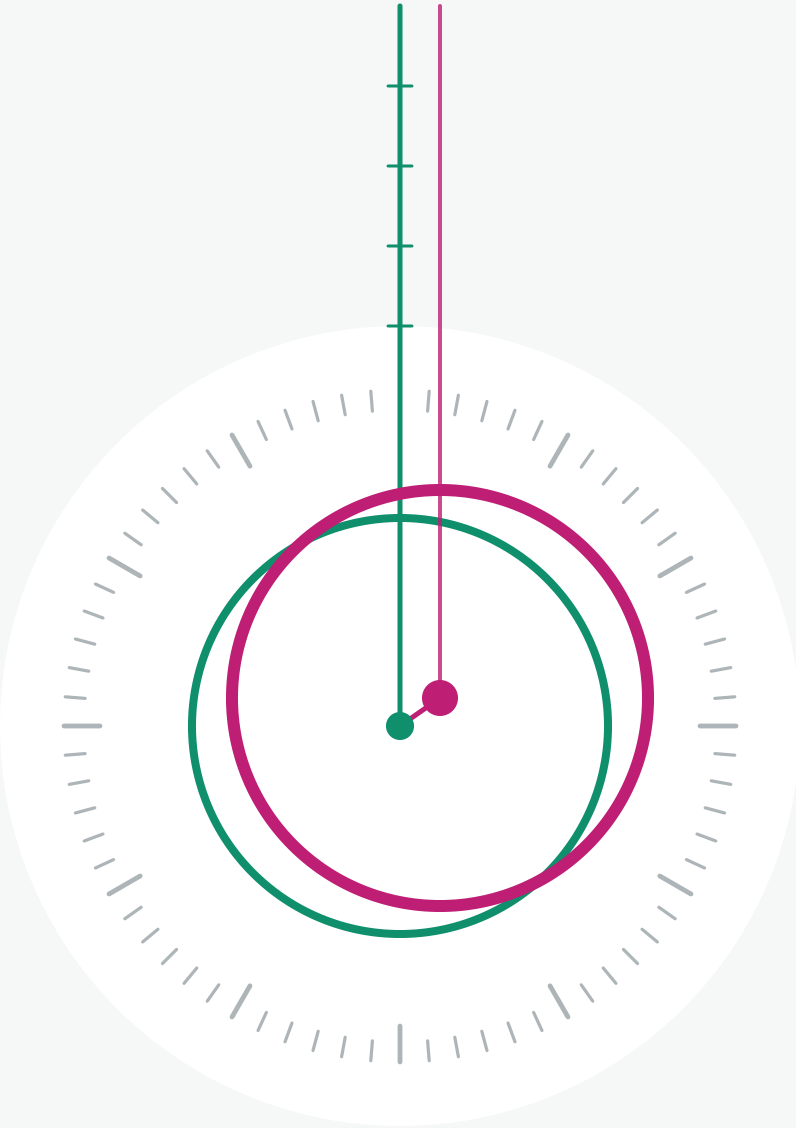
DATUM LINE

Confidence with controls.

Catch drifting cells before they cost a screen.



Kat Titterton, San Francisco



macrofluidic.me/resume

8
peer-reviewed papers
two in Nature Communications

2
patent applications
Axiom Bio, bioactivity detection

100,000+
compounds imaged
two Cell Painting datasets

Kat Titterton

2023–2026 **Axiom**
cell imaging, AI

2022–2023 **insitro**
automated screening

2021–2022 **MIT**
expansion microscopy

2018–2021 **abbvie**
neuron assays

2017–2018 **MGH**
electrophysiology

2017 **COLGATE** BA
neuroscience, math

MY WORK

1 Cell Painting in primary human hepatocytes
Cell Systems 2026, Ewald, Broad Institute

2 Counting cells predicts bioactivity benchmarks
Nat Commun 2026, Seal, Univ. of Cambridge

3 Cell Painting of hepatotoxic botanicals
bioRxiv 2026, Ollitrault, EMBL-EBI

4 An antibody to pathological tau
Mol Neurodegen Adv 2026, Yanamandra, AbbVie

5 Capturing synaptic proteopathic seeds
iScience 2026, Sebastian, AbbVie

6 A neuronal tau aggregation assay
J Biol Chem 2025, Wu, AbbVie

7 Multiplexed imaging of brain nanostructures
Nat Commun 2024, Kang, MIT

8 Tau aggregation in human iPSC neurons
iScience 2022, Manos, AbbVie

9 Neurosteroid photolabeling of $GABA_A$ receptors
Eur J Med Chem 2019, Wu, U Illinois Chicago

Do my cells look “normal”?

asked @ five benches
… answered @ none

“Normal” compared with whom?
Each lab’s own checks can pass while labs still disagree (slides 03 to 05, A13).

Each lab judges “normal” alone.
A wrong call costs $100,000s.

$28B a year of US lab research others cannot reproduce [6]

36% ≈ $10B, cells and reagents | 64%, other causes

search

U2OS cells “normal” morphology



Doil 2009 | Gustafsdottir 2013 | Stadler 2012

Rohban 2017 | Bray 2017 | JUMP 2023

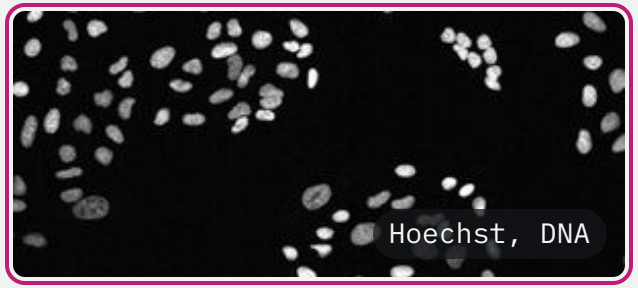
six papers, six “normals”

negative-control wells, IDR idr0010, idr0036, idr0025, idr0033, idr0016 (results/search-tiles.json); JUMP [7]; Doil and Rohban CC BY 4.0, Gustafsdottir and Bray CC0, Stadler CC BY-SA 3.0

ask

To: authors, U2OS paper

Subject: do my U2OS cells look right?



screenshot.png

one opinion

guess

RUN IT

$100,000s

at risk

about $50k a run, 3 runs a week, in the author’s experience [28]

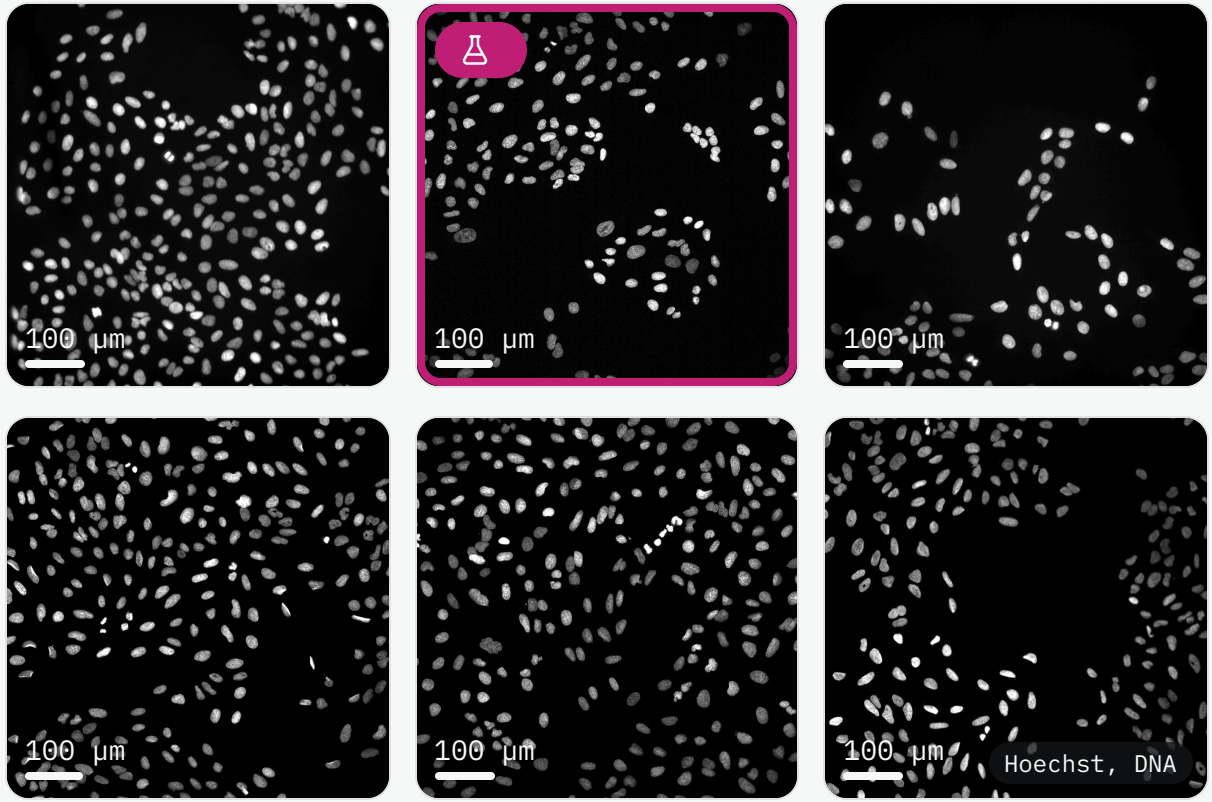
WAIT

months

lost

asking or re-running to be sure, the author’s estimate [28]

One lab, or six?



same cells

U2OS: human bone cancer cells

osteosarcoma (bone cancer) from a 15-year-old's tibia, described in 1967 [26]

the one line in JUMP's full dataset, so every lab here grew it [27]

chosen over A549 by a slim margin, for its public data [27]

named in 11 of the Cell Painting Gallery's 28 datasets [17]

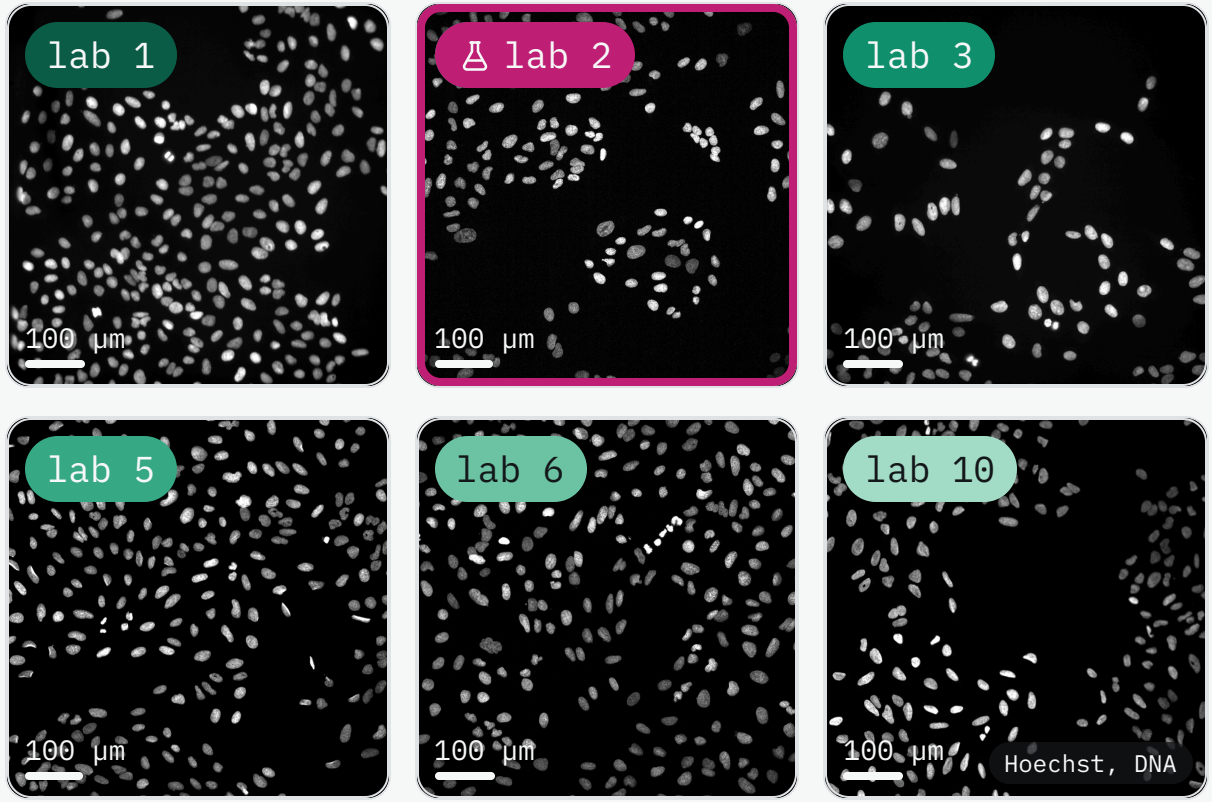
a cancer line: "normal" means its DMSO baseline, not healthy tissue

same protocol

no drug

fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

Could anyone tell?
A classifier can.

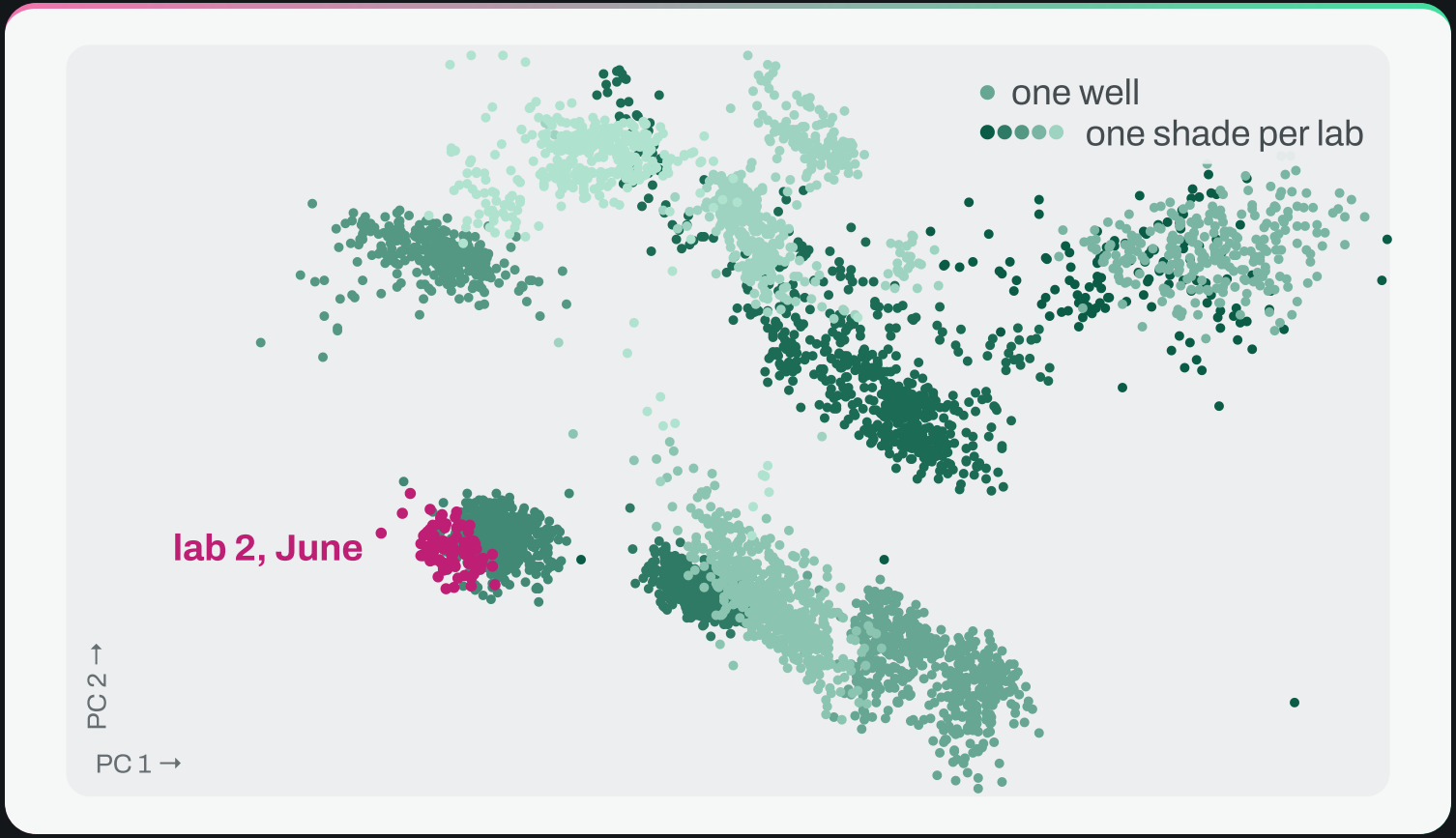


99.9% right lab, of 11

9% chance

fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

Eleven labs.
Eleven “normals”.



one well → 3,773 measurements → one point

size · shape · brightness · texture

6× further apart

lab to lab

spread inside one plate

6.2: the mean distance between two labs’ centres, in units of a well’s mean distance from its own plate’s centre; all 3,773 numbers (results/e13)

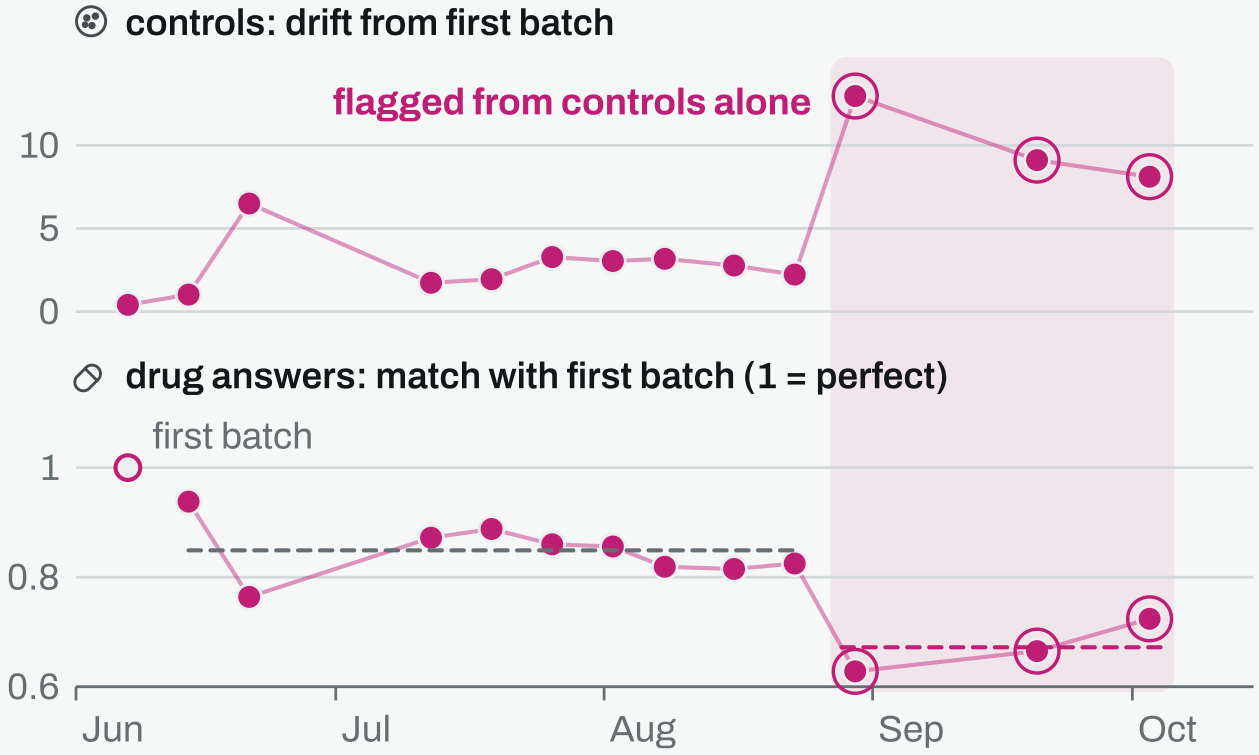
well one dish of cells

plate grid of wells

lab one site

fig 2 dots: DMSO wells, 8 per plate, one in three drawn; PCA of 15,352 wells, for the eye (distances use all 3,773 numbers); plates: 384 wells, 1,536 in labs 1 and 9; strip: a lab 2 DMSO well, Hoechst, DNA; JUMP [7]; A2, A3, A13

Controls alone flagged the batches whose drug results shifted.



Lab 2

By autumn it had drifted. From 30 Aug to 3 Oct, 3–5× its summer level (results/e10).

0.85 → 0.67

--- before --- flagged

match with the first batch: nine batches 14 Jun to 23 Aug, three flagged 30 Aug to 3 Oct

controls drift → drug answers shift

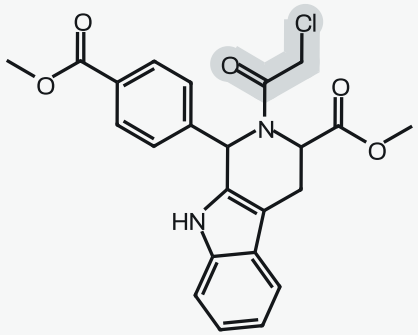
across eleven labs: ρ = −0.71, 118 batches, each lab centred (e10, A6)

batch: one run of plates

fig 3 lab 2, 13 batches; drift: control wells' distance from the first batch, in within-plate spreads; match: seven known drugs, 5 µM; dashed: means; flagged: most plates nearer another lab than the first batch (e9); A6; JUMP [7]

Even known drugs look different across labs.

RSL3



$C_{23}H_{21}ClN_2O_5$, as JUMP records it, no stereo; active form (1S,3R) [19]

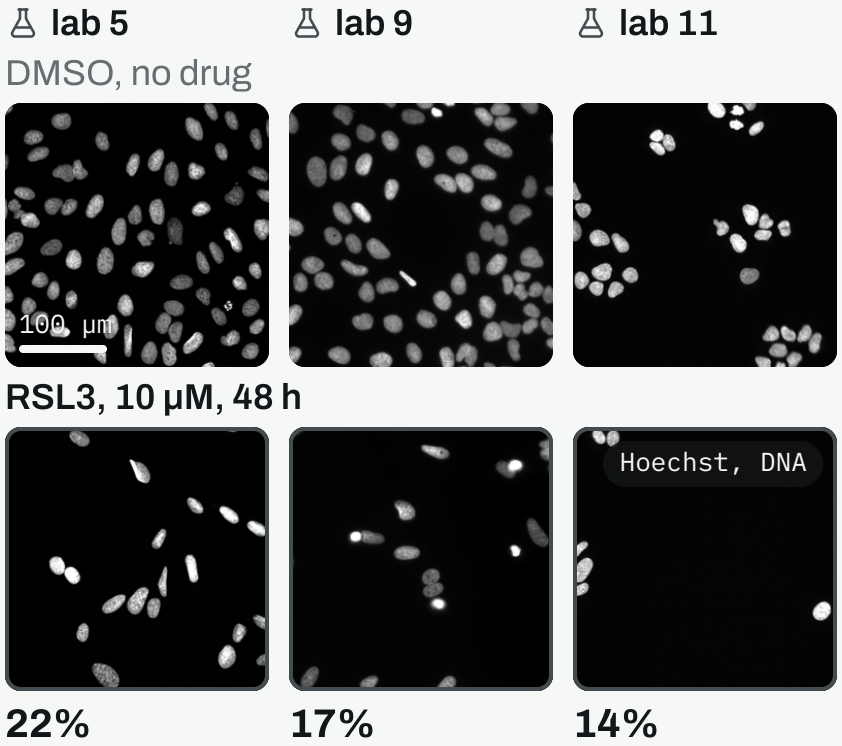
shaded group binds GPX4 [19,21]

↓

lipid peroxides pile up [20,22]

↓

cell dies ferroptosis [20]



14–22%

cells left

same dose, same kill

same look

same lab

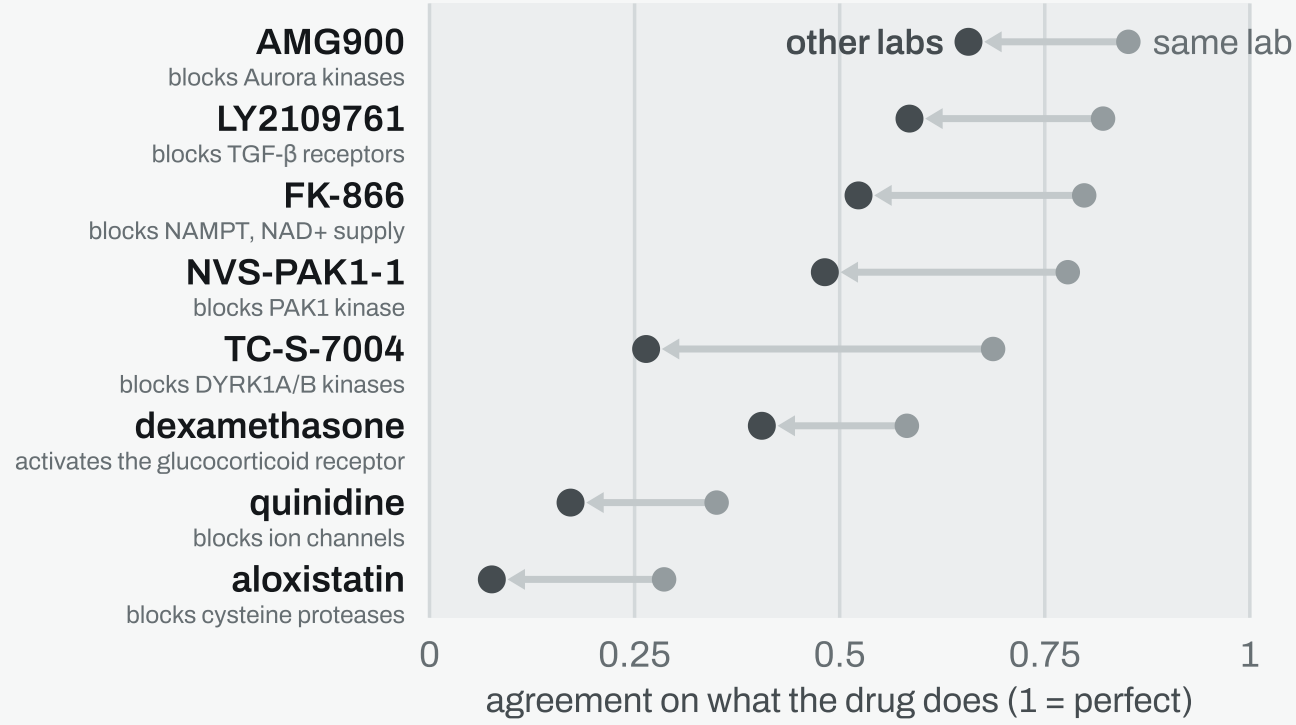
other labs

Bars: agreement well to well at 10 µM, 0.90 in the same lab, 0.49 across labs (1 = perfect); known drugs on the same plates 0.74 → 0.26.
An example, not proof: seven wells in three labs (A9).

fig 4a RSL3 at JUMP's dose [7], one well per plate; cells left: nuclei per field against the same plate's DMSO wells, lab means; lab 7 (0.625 µM, 80% left) left out, A9; 1,200+ PubMed records on RSL3 [16]; it blocks TXNRD1, not recombinant GPX4 [22]

Even known drugs look different across labs.

Eight known drugs, 5 µM in every lab.



WHAT IT COSTS

- hits don't replicate [13], e8
- data won't pool 06, [13]
- models learn the lab 03, e1

mean agreement between plates: same lab 0.64 → other labs 0.40

ON THE MAP

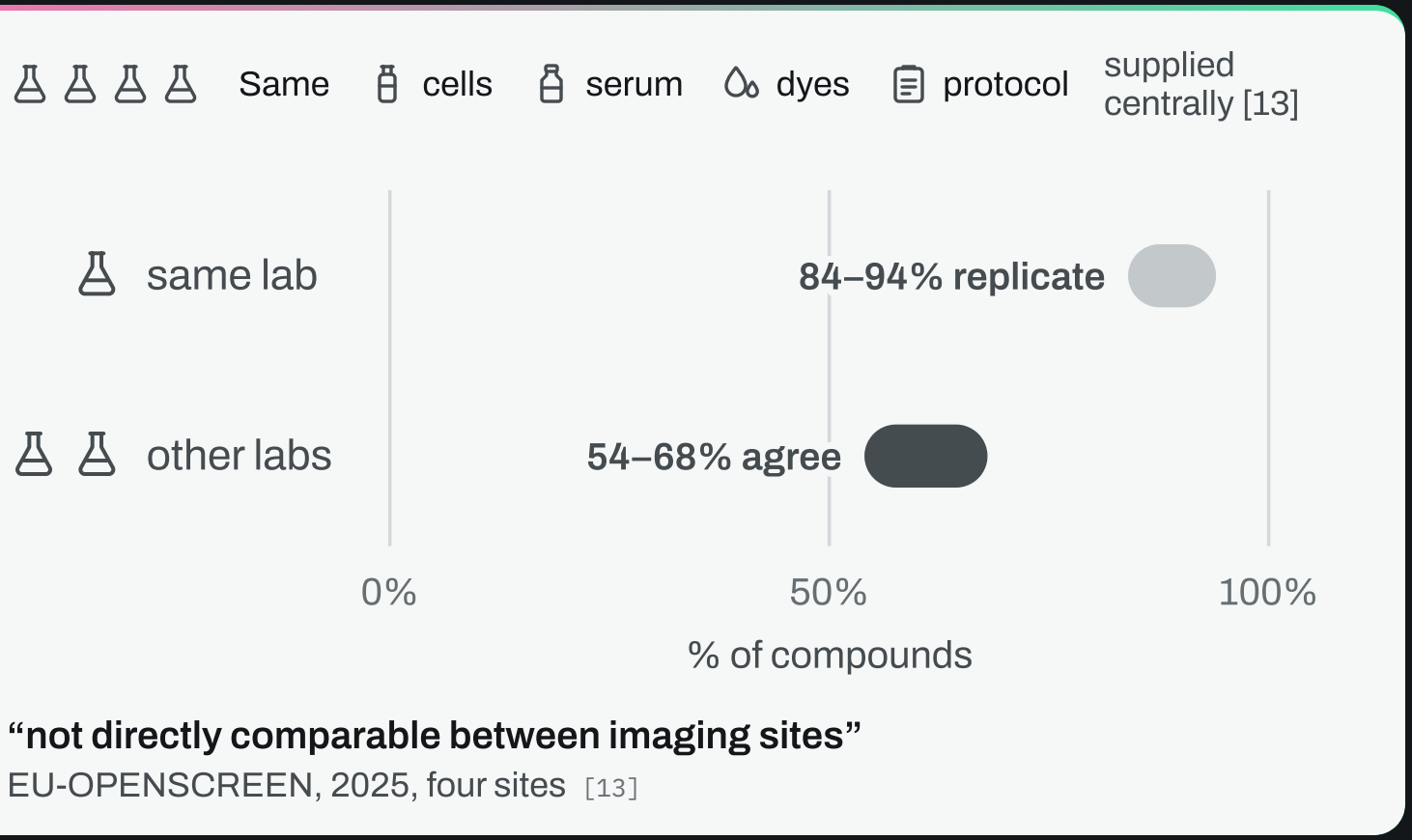
107 of 125

batches agree more with other labs

than on their own controls
by chance: about 62

fig 4b plate pairs, same lab and across labs, 5 µM [7], 1,829 plates, 11 labs (LY2109761 in 10), targets [24]; per batch, 7 drugs labs agree on (aloxistatin 0.38, under 0.5, e10): 0.75 raw → 0.76 usual fix → 0.78 on the map; A9, A14, results/e8, e12

Cell data is worth millions.
Only if labs can pool it.



“not directly comparable between imaging sites”
EU-OPENSCREEN, 2025, four sites [13]

fig 5 EU-OPENSCREEN [13], two measures: same lab, percent replicating (Hep G2 at four sites, U2OS at one, filtered); other labs, percent pairing (Hep G2); A23

$30M

one cell dataset

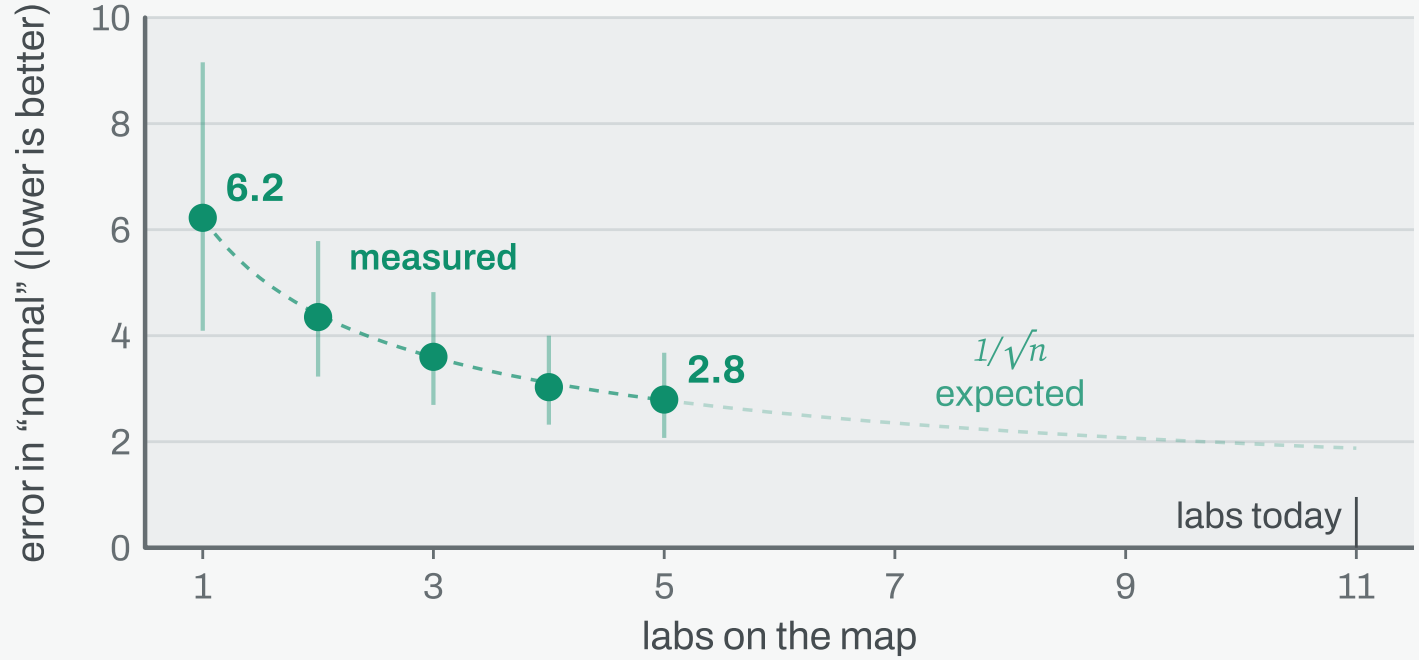
Roche paid it: a milestone for one whole-genome map of microglia [11]

Why now

2023	1.8 billion cells, public [7, 25]
2025	FDA moves off animal tests [9]
now	AI reads cell images, A10

JUMP went public from 2023 [7] and counts 1,810,840,525 cells today; source_15 is not counted, so these are not this map’s eleven labs [25]

Every lab that joins sharpens “normal” for everyone.



6.2 → 2.8

error in “normal”: one lab, then five

Map, today

↓

Certificate, next

↓

Subscriptions, then

↓

Journals, raw images, every paper

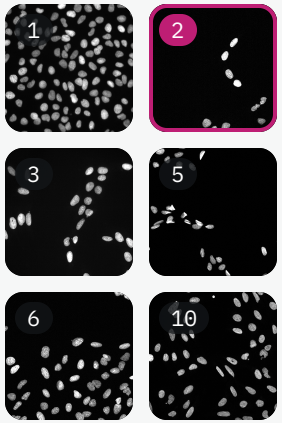
↓

…

fig 6 error: distance between two maps from different labs, in within-plate spreads; dot: mean of 400 draws, bar: 10th to 90th percentile, measured to five labs, dashed beyond, 1,871 plates, JUMP [7], results/e13

One shared repository of raw cell images.

free



Hoechst, DNA

eleven labs, control images (A18)

DATUM LINE

- **one map** every lab placed on it
- **raw images** kept, with their provenance
- **certificate** every batch read against the field
- **benchmark** every method graded on one test

subscription

- **scientists** onboard assays with confidence: is my batch "normal"?
- **AI agents** query comparable data, by API
- **companies** drug discovery, stronger together: a certificate on data they buy

Proposed terms: labs keep their data and a share of every subscription. A15, A24.

93,228

control wells · 11 labs

93,196 of them on the map; one plate has no batch (results/e1, e13)

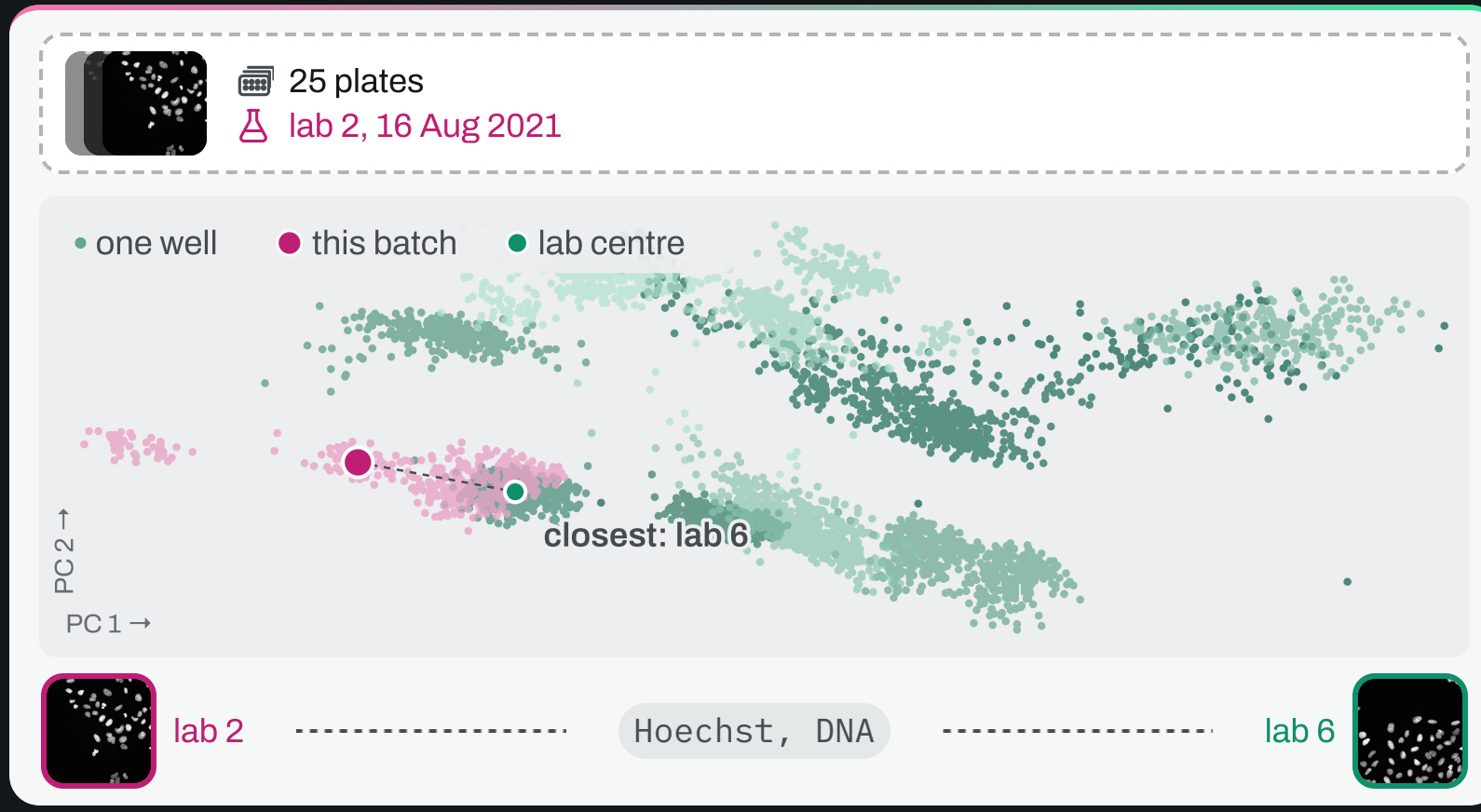
One subscription: the repository, the map, the benchmark.

Public archives keep finished datasets (656 TB, free [17]). This keeps every lab's live controls, aligned and checked.

Every cell result becomes checkable against the field.

the product as proposed, counts from results/e1 and e13, JUMP [7], A15, A18, A24

Do my cells look “normal”, next to every other lab?



Certificate

Do my controls look “normal”?

Do my drugs match other labs?

known drugs worked; agreement with other labs: 0.70 measured, **0.77** on the map (1 = same)

Did my cells grow as usual?

in range: 188 cells per well

fig 7 a proposed certificate, replayed on public data; closest lab in all 3,773 numbers: lab 6, 4.4 within-plate spreads away; JUMP [7], results/e9, e11, e12; A11, A18

Which labs make the most reliable data? Ranked on controls alone.

Ranked on controls alone

off centre, spread, drift from last batch, cell count; U2OS, A17

rank			score, higher is better	
#1	lab 5	28 Jun		82
…				
#48	**lab 2**	2 Aug	best	**58**
…				
#95	**lab 2**	16 Aug	this batch	**36**

certified in a row ✓✓✓✓✓✓✓✓✓✓ lab 2's streak; longest in the field: 33

Standing

#95 of 129 batches

Batches certified

10 of 13

To climb: steadier batches

Lab 2's drift from last batch is more than twice the field's.

2.9

1.2

drift from last batch, lower is better: lab 2, then the field's median batch

private, unless the lab shares it

fig 8 score = 100 − mean percentile of four measures, DMSO wells, every batch: 129 (05b counts 125: it leaves out the four lab 3 batches whose known drugs failed, e11, e12); JUMP [7]; A17

Six questions, six tools.
Use all of them.

Each answers one question well.
Each misses something, Datum Line too.

tool	answers	misses
DNA fingerprint (STR)	Is it the line the label says?	how the cells behave A7
vendor certificate	Was it fine when it shipped?	anything after shipping
plate controls	Did this plate work?	a whole lab drifting A1
in-house platforms	Is my own data consistent?	lab differences; it removes them A3
data factories	Where do I get cell data at scale?	other labs' "normal" A23
Datum Line	**Do my cells look "normal", next to every other lab?**	which cell line it is; unstained images, not yet A7, A10

STR: ATCC [14]

Eighteen months to first revenue.

WHAT THE MONEY BUYS

$1.8M

pre-seed, $0.9M line by line, ×2 for the usual overrun, A21

→ seed round

1

	widefield	confocal
U2OS	■	■
A549	☐	☐
HepG2	☐	☐
iPSC	☐	☐

benchmark playground

known drugs, planted faults, A4, A16

2

11 + 3

labs today + 3 partners

three partner labs on the map

U2OS first, then any line

3



99.5%

right lab of 8

model for any image

from one brightfield image; chance 12.5%, A10, A19

4

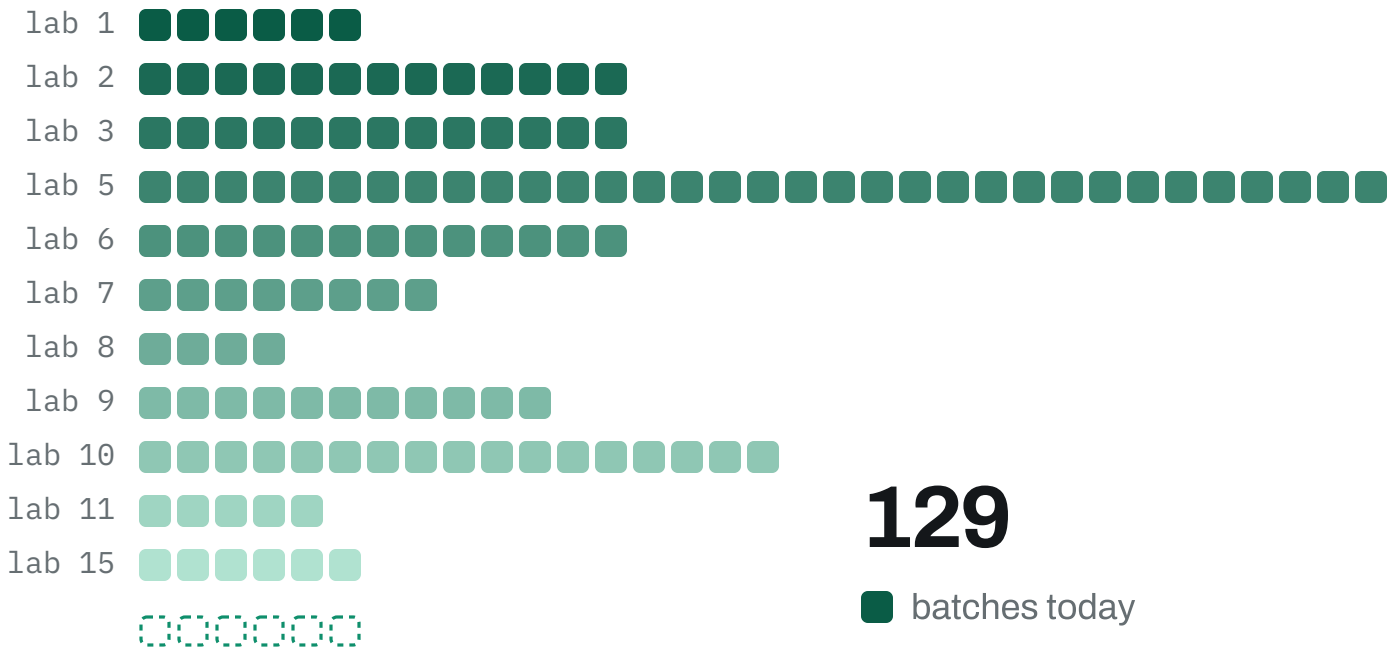
first paid certificate

a company pays ~$10–50k per check, A15

proposed, milestones in order, not dated

Every plate imaged, placed on the map.

Labs share control images, free.
Each sees its “normal” next to everyone else’s.



$1.8M

pre-seed, eighteen months, A21

1 **leader**, to run it

3 **partner labs**, bench time

2 **benchmark partners**

**Have control images?
Talk to me.**

fig 10 one square: one batch, a median of 10 plates (1 to 58); 1,871 plates, JUMP [7], results/e9; A15, A20, A22, A24

The rest, for questions.

Every number comes from the code here or a cited source.
Unfinished work is marked pending.

A1 **Why not trust the plate's controls?**
sick plate, sick control

A2 **What is the math?**
six calculations on 3,773 numbers

A3 **Why not batch-correct it away?**
erases most of the lab fingerprint

A4 **Is the model the moat?**
counting cells often wins

A5 **Why not just count cells?**
same count, different texture

A6 **How often do I re-certify?**
your "normal" expires

A7 **Isn't this cell-line DNA testing (STR)?**
which cells, how they behave

A8 **Is the assay itself working?**
known answer

A9 **Does it change the result?**
same drug, different answer

A10 **Does it work without stains?**
bet on brightfield

A11 **Does it flag too often?**
rarely, six of 129 batches

A12 **What moved?**
stain brightness, not cell count

A13 **How does one lab move over time?**
own "normal" fine, other labs: moved

A14 **What does the map buy?**
same drug, closer between labs

A15 **Who pays?**
labs join free, subscribers pay

A16 **What would the money build?**
ugliest dataset, on purpose

A17 **Is the leaderboard fair?**
ranked on controls, never on results

A18 **What is private?**
control wells leave, nothing else does

A19 **What is the machine learning?**
image in, position and correction out

A20 **What about biosecurity?**
cells show their lab, labels can lie

A21 **How much, and what for?**
planned $0.9M × 2 = $1.8M asked

A22 **How big?**
count × price, three streams

A23 **What is cell data worth?**
priced in deals, labs disagree

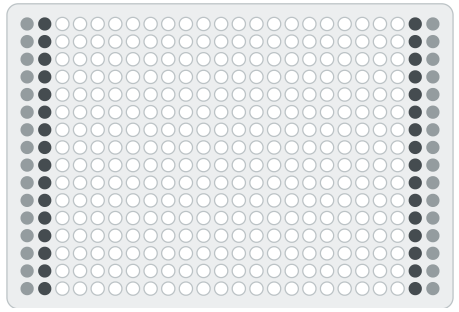
A24 **Why would a lab share?**
owned by the lab, verified by anyone

A25 **What does the shared data look like?**
one dataset per lab, certificate attached

If the plate is sick, its controls are sick too.

Controls sit on the same plate as the tests.

A REAL JUMP PLATE, 384 WELLS



DMSO 32 known effect 32
test 320



test = *"normal"* + *plate* + *effect*

control = *"normal"* + *plate*

difference = *effect on this plate*

plate cancels from the number, never from the cells

WHAT CAN SEE A SICK PLATE?

A view from outside the plate.
Same cells in other labs: the map.

Fig. A1 left: JUMP plate 1053601756, lab 2, right: schematic, no measurements

Every result is one of six plain calculations.

Results compare averages, or test a model on unseen plates.
Drug discovery removes the lab; a quality check reads it.

1, IMAGE TO NUMBERS [1]

$$e = \varphi(image)$$

Three ways to turn images into numbers gave similar results on liver cells.
Tool is not the asset.

2, WHICH LAB? 99.9%

$$f : e \to lab$$

Names the lab from one well, on plates it never saw.
Far above chance (9%): each lab leaves a fingerprint.

3, HOW FAR FROM "NORMAL"? [e6]

$$d = \| E[e \mid plate] - E[e \mid all\ labs,\ c] \|_{\sigma}$$

How far this plate's average sits from all labs' average, in units of usual spread.
Direction says what moved.

4, SOURCES OF VARIATION pending

$$Var(e) = V_{model} + V_{lab} + V_{batch} + V_{plate} + V_{well}$$

Splits variation into microscope type, lab, batch, plate, well.
Shares add up to 100%.

5, REMOVE KNOWN CAUSES pending

$$e_{\perp} = e - \hat{E}[e \mid c]$$

Remove only causes known to be the instrument.
Subtracting each plate's controls goes further: right lab only 24% of the time.

6, DID THE DRUG CHANGE THE CELLS?

DISTANCE $\delta = b(\|e - E[e \mid DMSO]\|)$

CLASSIFIER $g : e' \to label$

Distance from the DMSO average, capped so one odd well cannot dominate.
Or a model, after aligning to the reference.

The usual correction erases most of the lab's fingerprint.

What counts as "normal" depends on every variable below.
JUMP records the left column; nobody records the middle.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

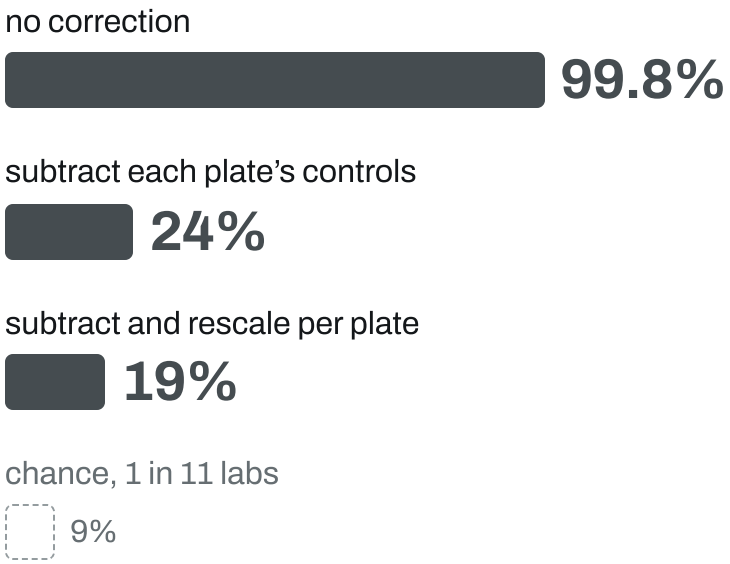
in every image file too:
image size, stored bit depth

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity
- compound dose

not required to publish today
in our own data: tracked, every one

CAN A MODEL TELL WHICH LAB?
HIGHER = CLEARER LAB FINGERPRINT



29,943 DMSO wells, 16 per plate
tested on plates it never saw

Batch correction is built to remove this signal [18], variables: JUMP metadata [7]

nature communications, 2026, [2]

On weak benchmarks, counting cells often wins.

I coauthored this paper.
Benchmarks decide which model looks good.

… counting cells achieves unexpectedly high performance in these benchmarks, making them less useful for discerning whether additional properties, such as phenotypic profiles (mRNA or Cell Painting), provide additional useful information on bioactivity. … we recommend filtering benchmarks to exclude such assays and including a cell-count baseline. … we confirm that models leveraging Cell Painting image-based profiles outperformed the baseline cell count model. …

ABSTRACT EXCERPT, VERBATIM finding recommendation, what held up

209 and 270

tests in two popular benchmarks
many really measure cell health

Images win

on 24 tests aimed at one protein each
Cell Painting beat the cell count

WHY IT MATTERS HERE

Whoever sets the baseline sets the benchmark.

benchmark playground, a cell-count baseline on every task, planted faults, every model scored, A16

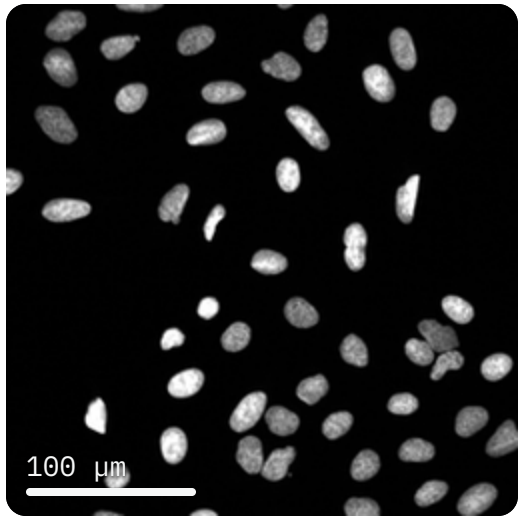
Seal S, Dee W, … **Titterton K**, … Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*, 2026.

Same cell count, different texture: only images see it.

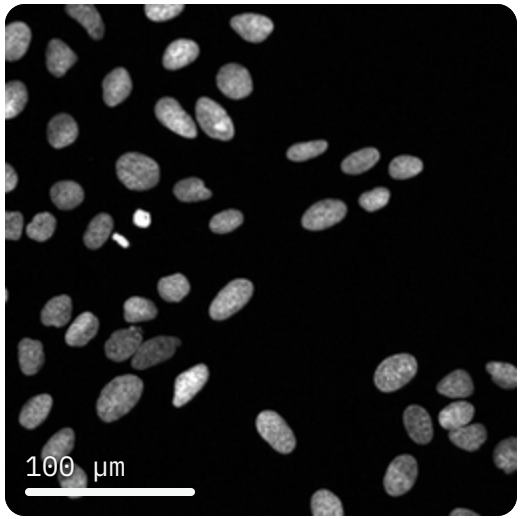
Count moves only when cells die or divide.
Image moves when anything changes.

SAME CELL LINE, SAME PROTOCOL, SAME MICROSCOPE

each lab's typical plate



LAB 5 CV8000, confocal



LAB 6 CV8000, confocal

CELLS COUNTED

196 ≈ 222

gap 0.4× the usual spread → same

FINE DETAIL, % OF DNA SIGNAL

0.94% ≠ 1.10%

gap 2.4× the usual spread → different

fine detail: finer than 5 µm, inside nuclei
usual spread: plate to plate, within one lab

NOT ONE LUCKY PAIR

All 44,325 plate pairs, lab 5 × lab 6:
texture differs in 91%, count in 33%.

TWO WAYS TO READ IT, A2

Classifier: does it belong with past runs?
Distance: how far out, which way back?

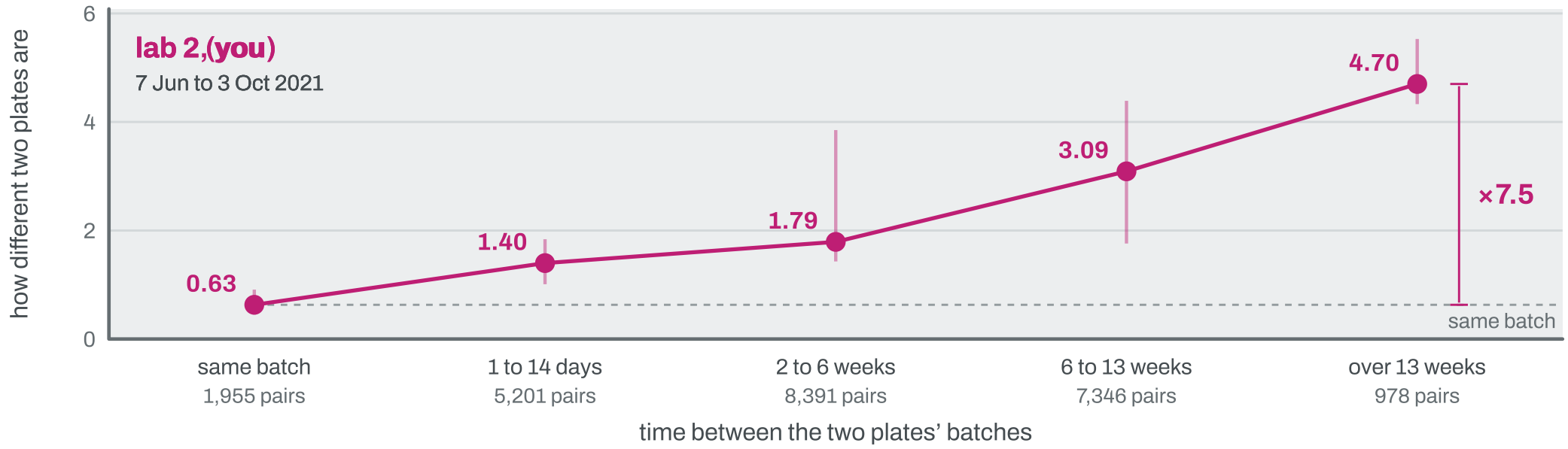
BOTH NEED A REFERENCE

Somebody else's images.
That is the map.

Fig. A5 DNA channel, 300 µm squares, JUMP cpg0016 [7], CC0, nuclei by Otsu, both ways: published [1], our tuning unpublished

Plates drift within two weeks: re-check every batch.

Two plates from one batch differ by 0.63; two weeks apart, 1.4. The jump past 13 weeks is lab 2's autumn move (A13).



IS THE CELL LINE RUINED?

This data shows when, not why.
What moved in lab 2: A12.

HOW FAST?

Within 2 weeks: ×2.2 vs same batch.
6 to 13 weeks: ×4.9.

HOW SOLID IS THE NUMBER?

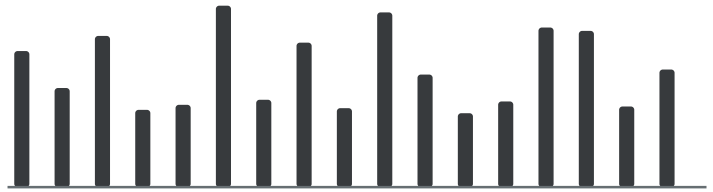
One lab, 219 plates, 13 batches.
Other labs not yet measured.

dot = median pair of plates, bar = middle half of pairs, 1 = how far a typical DMSO well sits from its plate's average, JUMP [7]

STR says which line; images say where it was made.

One cell line, U2OS, in all 11 labs.
Images still name the lab: 99.9% right.

STR, DNA FINGERPRINT

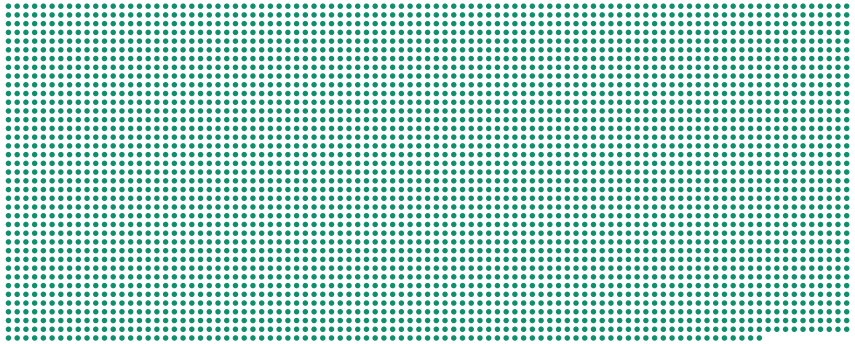


17
DNA markers

periodic
per cell line [14]

Standard test for identity.
Cannot say where or how the cells were grown.

DATUM LINE



3,773
numbers per well

every batch
images already taken

Shows where cells were made, next to other labs.
Cannot say which cell line it is.

identity, STR + behaviour, Datum Line = **one certificate**

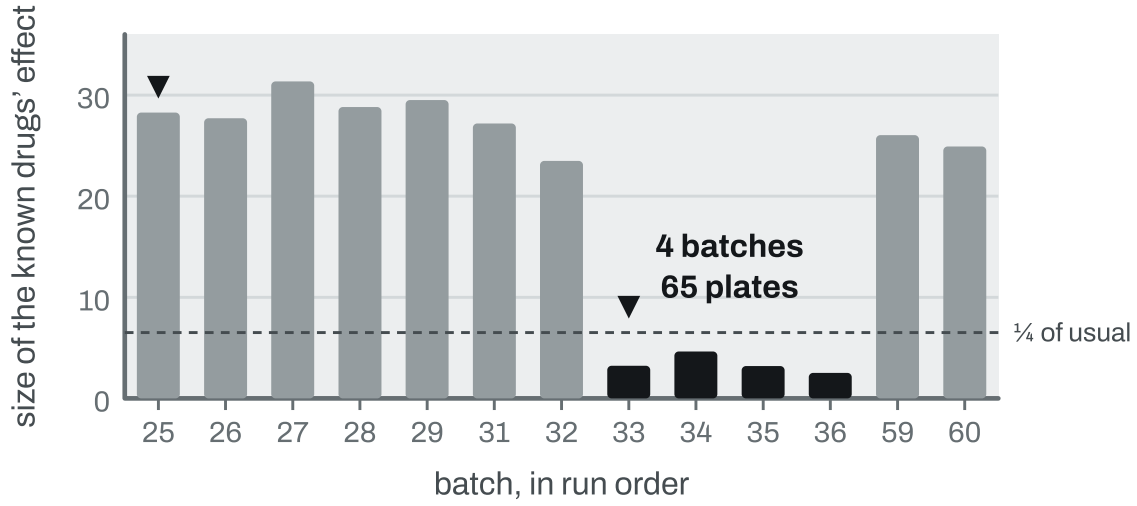
natural first partner: STR testing services

STR: 17 loci + amelogenin, ATCC [14], Datum Line: 93,228 U2OS DMSO wells, 1,872 plates, 11 labs, plates held out, JUMP [7]

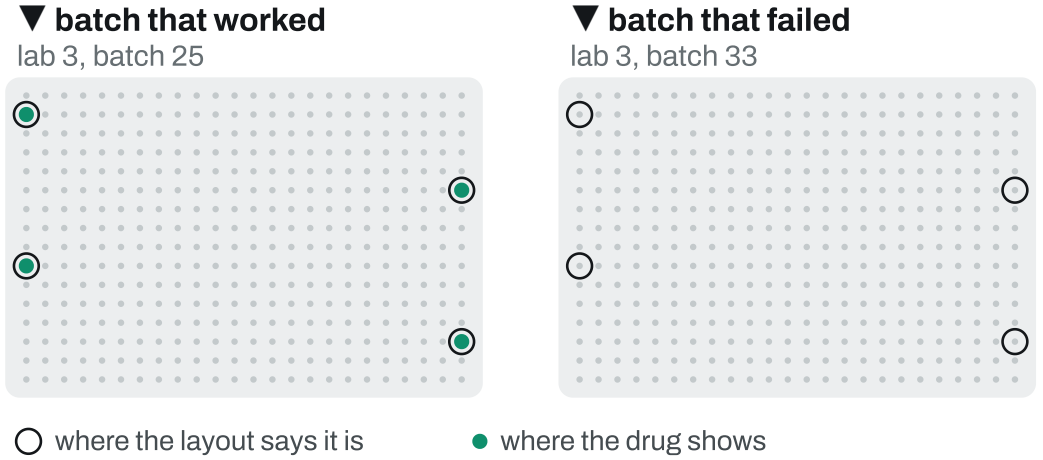
Known drugs show whether the test itself worked.

DMSO wells show where a lab sits.
Lab 3: 4 batches where the known drugs did not work.

LAB 3, DID THE KNOWN DRUGS WORK?



WHERE DOES THE KNOWN DRUG SHOW?



125 of 129
PUBLIC BATCHES
KNOWN DRUGS WORKED

1 of 4
FAILED BATCHES CAUGHT BY
DMSO WELLS ALONE → USE BOTH

–0.71 → –0.70
FIG 3 LINK, DRIFT VS ANSWERS
WITHOUT THESE 4 BATCHES: HOLDS

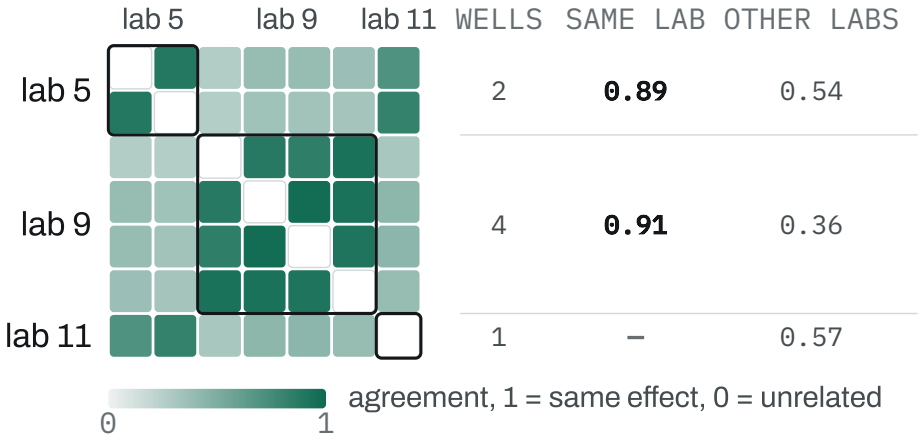
failed = under ¼ of the lab's usual effect + unlike other labs, 1,829 plates, maps: plates C13443aW, A12069aW, JUMP layout [7]

The same drug looks less alike in another lab.

RSL3 at 10 µM from fig 4a, every well against every other.
Darker = more alike.

WHAT A DRUG SCREEN REPORTS — drug wells – DMSO wells, same plate

RSL3 AT 10 µM, EVERY WELL AGAINST EVERY OTHER



	WELLS	SAME LAB	OTHER LABS
lab 5	2	**0.89**	0.54
lab 9	4	**0.91**	0.36
lab 11	1	–	0.57

SAME 7 PLATES, EACH WELL AGAINST ITS OWN PLATE'S DMSO

	SAME LAB	OTHER LABS
RSL3, 10 µM	**0.90**	0.49
8 known drugs, 5 µM, median	**0.74**	0.26
DMSO vs DMSO	**0.00**	0.00

DOES IT CHANGE THE NUMBER?

Which lab ran it explains 20% to 71% of how strong a drug's effect looks.

HOW SOLID?

Seven wells: lab 5 two, lab 9 four, lab 11 one. Lab 11 has no same-lab pair.
Most of the gap sits beside lab 9, on 1536-well plates: 0.33–0.39 against 0.75.
Its known drugs split the same way.

LAB 7, LEFT OUT

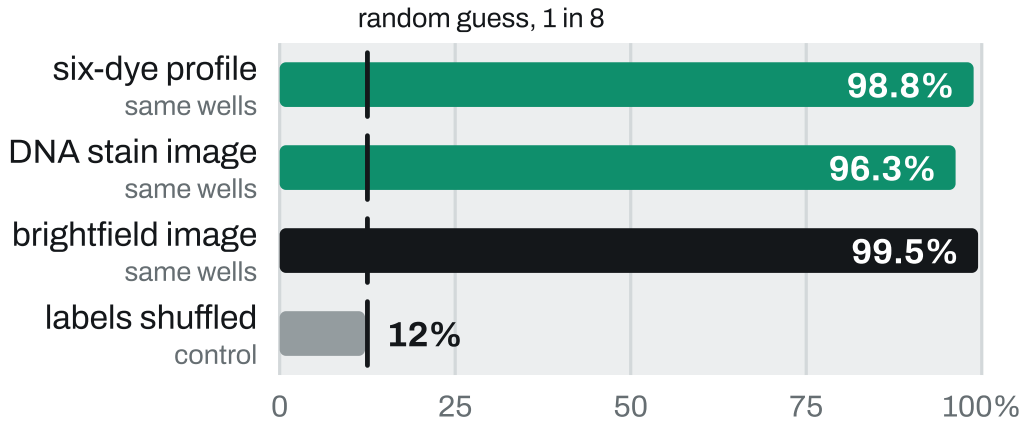
Dosed at 0.625 µM, a sixteenth [7]: RSL3 left 75–87% of its cells.
AMG900 at 5 µM left 36% there, 28–30% elsewhere: its cells respond.

agreement = cosine of two wells' effects, each well minus its own plate's DMSO wells, 8 drugs at full scale: 1,829 plates, 11 labs, JUMP [7]

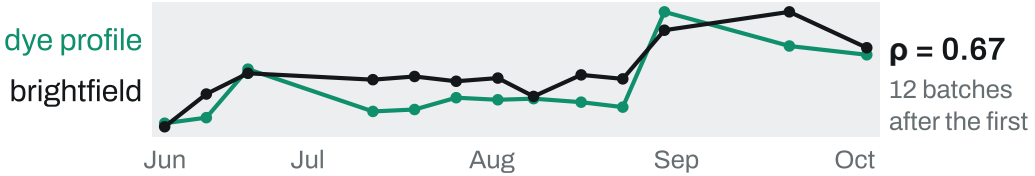
One brightfield image already names the lab.

Evidence: cells stained with six dyes.
Product: plain light, no dye.

CAN ONE WELL TELL WHICH OF 8 LABS IT CAME FROM?



DOES BRIGHTFIELD SEE LAB 2 DRIFT?, EACH LINE SCALED TO ITS PEAK

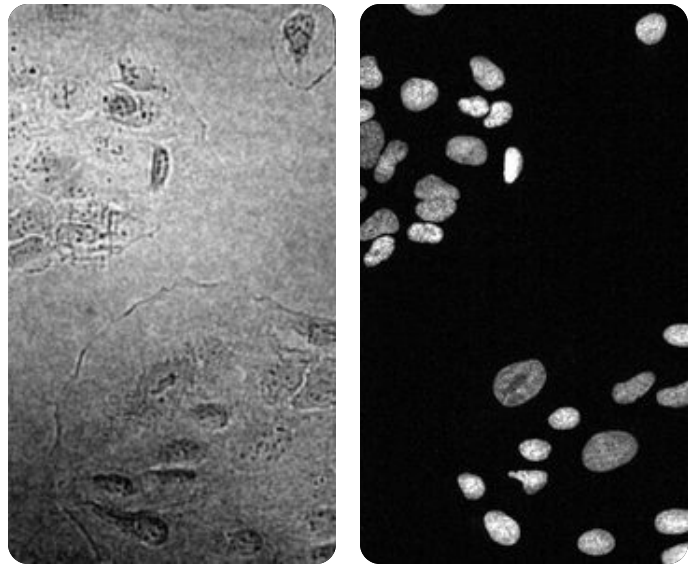


Not yet the whole map: beyond each lab's usual neighbour, brightfield and dye profile agree batch by batch no more than luck (45 of 102 against 41).

WHY THE BET IS FAIR

No dye: fewer things can go wrong

~~stain lot~~ ~~dye lot~~ ~~staining step~~ A3



BRIGHTFIELD DNA STAIN

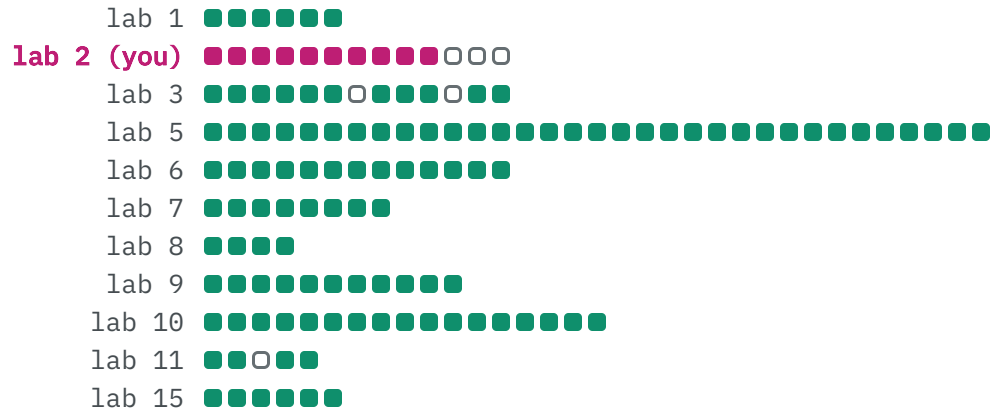
same spot in one well, lab 2

% named right, each lab counted equally, unseen batches, one DMSO well per plate, 1,338 plates, the same wells and model for every row, median of 20 splits of the batches, profile without its 8 frame-size columns, shuffled: 5 seeds per input, image model never trained on cells, JUMP [7]

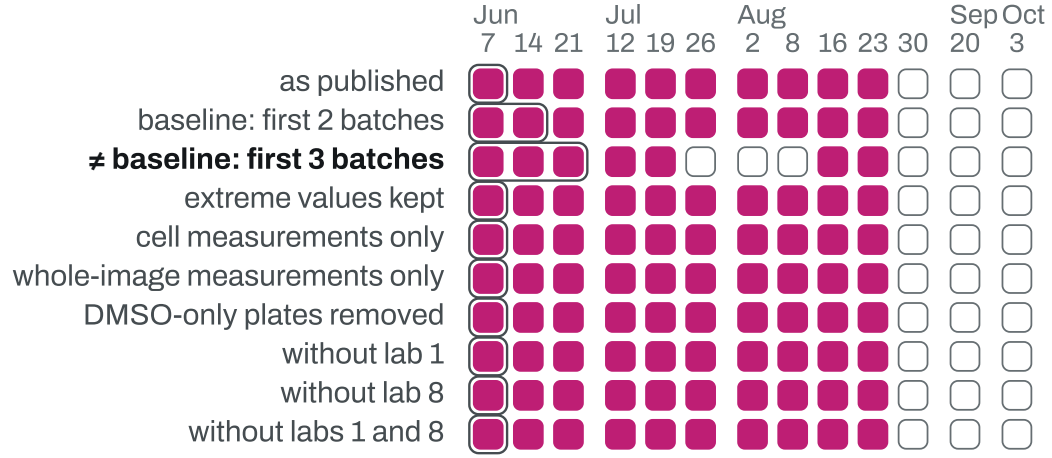
It flags rarely: six of 129 batches.

8 of 11 labs never flagged.
The one way that differs flags more, not less.

ALL 11 LABS, 129 BATCHES



LAB 2 (YOU), 13 BATCHES × 10 WAYS OF CHECKING



lab 2 (you) · other labs · flagged · baseline: the batches it is compared with

WHAT A FLAG MEANS

Batch nearer another lab than its own “normal”, its first batch

LAB 2 (YOU), PLATES FLAGGED

25 of 25 in flagged batches

2 of 194 in the other batches

THE ONE WAY THAT DIFFERS

+3 flags, 26 Jul → 8 Aug

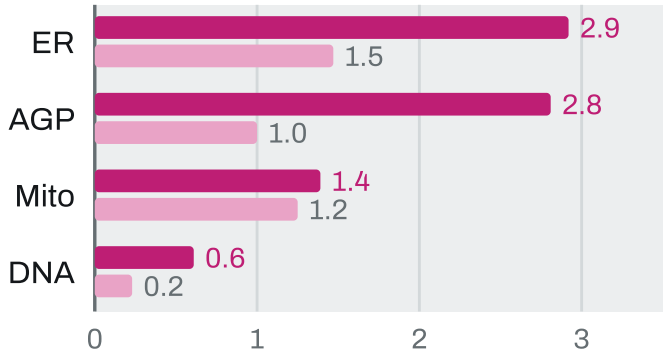
Its baseline takes in 21 Jun:
150 cells per well, lab 2’s lowest, A13

Each plate checked against its own lab’s first batch, 1,871 plates, 11 labs, JUMP [7], lab 2 over time: A13

Stain brightness rose; the cell count did not.

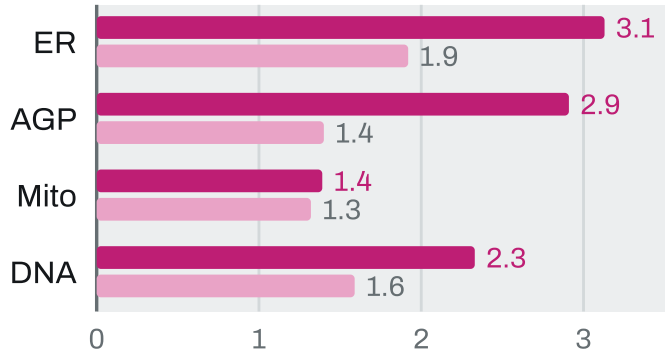
Lab 2 (you): flagged batches against its own “normal”, its first batch (7 Jun).

BRIGHTNESS



shift from 7 Jun, 0 = none, 1 = usual spread across labs

PATTERN (TEXTURE)

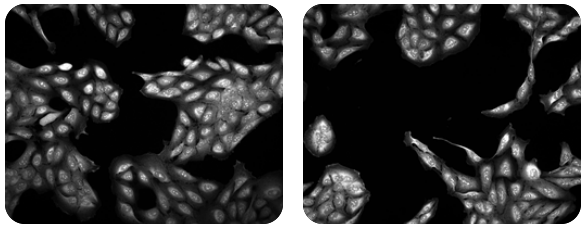


size of shift from 7 Jun, same scale

30 Aug, first flagged, 5 plates

3 Oct, latest, 14 plates

ER, BOTH LOOK FINE ON SCREEN



7 JUN

30 AUG

7× brighter in raw camera counts, one field each
7,612 → 51,230

one field each, ER pattern: around the nucleus → whole cell

LAB 2 (YOU), IN NUMBERS

ER + AGP brightness: up in ≥ 96% of measurements
cells per well: 189 → 165 → 182
inside its own range, 150–199, A13

CAMERA OR CELLS?

Cell measurements moved,
not only whole-image ones.
New stain batch or real biology?
This check cannot tell.

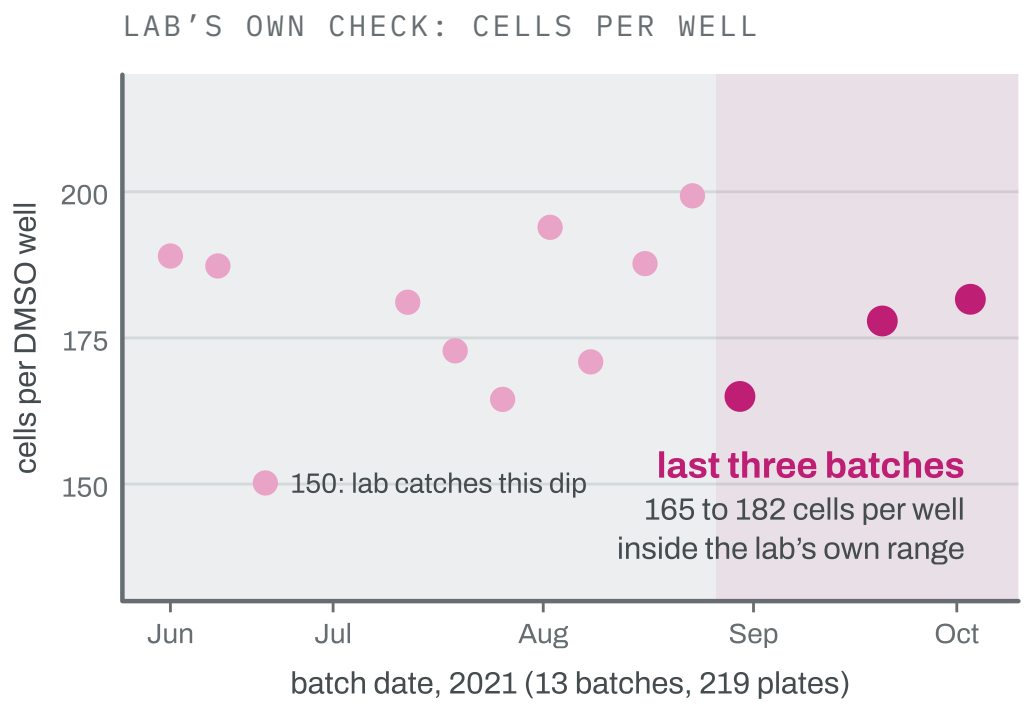
WHAT YOU GET

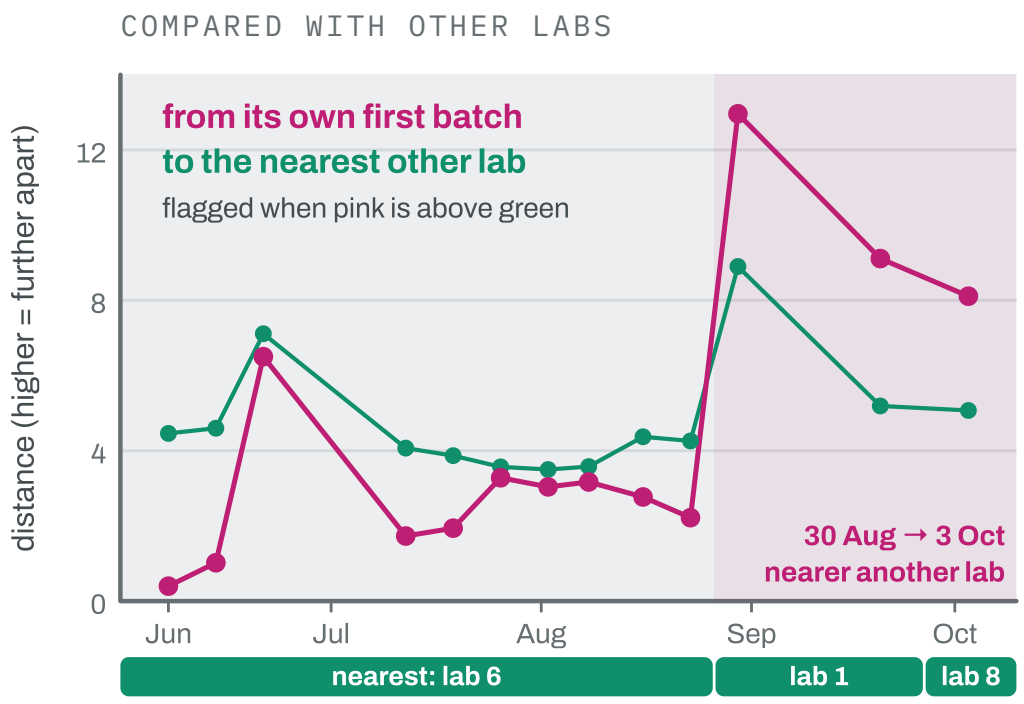
Look first: camera exposure, ER stain
Then: stain mix-up, colour leak, biology
Your lab finds the cause

ER, AGP, Mito, DNA: four stain channels, each marking different parts of the cell, each bar averages one channel’s measurements, JUMP [7]

By its own count lab 2 was fine; beside other labs, it had moved.

Lab 2, 13 batches, two views.
Not a verdict: a reason to look.





distance unit: usual spread between wells on one plate, lab 2, 2021, JUMP [7], which batches: A11, what moved: A12

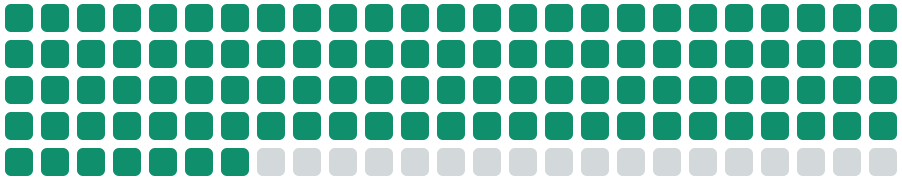
On the map, labs agree a little more on the same drug.

Seven drugs with a known effect, 125 batches, 11 labs.
Higher is better.

Do labs agree on what a drug does?

0.75 raw → **0.76** usual fix → **0.78** on the map

1 = every lab sees the same effect

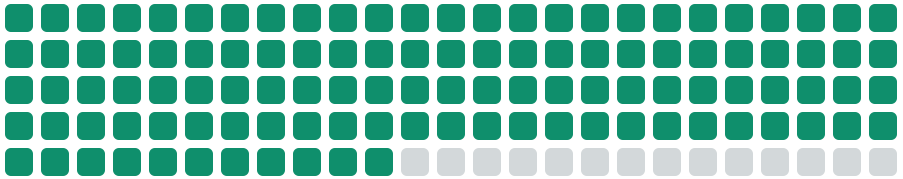


107 of 125 batches higher on the map than with the usual fix
than raw: 121 of 125

Can you still tell the drugs apart?

0.52 raw → **0.56** usual fix → **0.59** on the map

gap to the nearest wrong drug, higher is easier



111 of 125 batches higher on the map than with the usual fix
than raw: 125 of 125

NOTHING LOST

Right drug recognised: 0.92 → 0.94
Matches its own past: 0.88 → 0.88

HOLDS

Labs agree more: 5 of 5 ways of checking
Drugs stay apart: 4 of 5

NOT FIXED YET, DRUG STRENGTH

Varies batch to batch, lower is better
raw 6% → usual fix 20% → map 12%: steadier than the usual fix, not yet as steady as raw

usual fix: each batch set against its own “normal”, its DMSO wells, map: against every other lab’s, 1,829 plates, JUMP [7]

Labs join free; subscribers pay for access.

Certificate pays first.
Then subscriptions to the repository: access, not ownership.

WHO	WHAT THEY GET	WHAT THEY PAY	WHAT IT BUILDS
labs and core facilities	a place on the map certificate, leaderboard	**free, for sharing controls**	the map
companies that buy cell data	a certificate on every dataset they buy	**~$10–50k per certificate**	first revenue
teams that run cell screens all year	a check on every batch they run	**~$5–15k per cell line, per quarter**	repeat revenue
AI labs and model builders	a test set with faults planted on purpose	**~$50–200k a year, as partners**	the standard
then: subscribers to the repository	access to certified raw images from many labs	**a subscription, each lab keeps a share**	the network

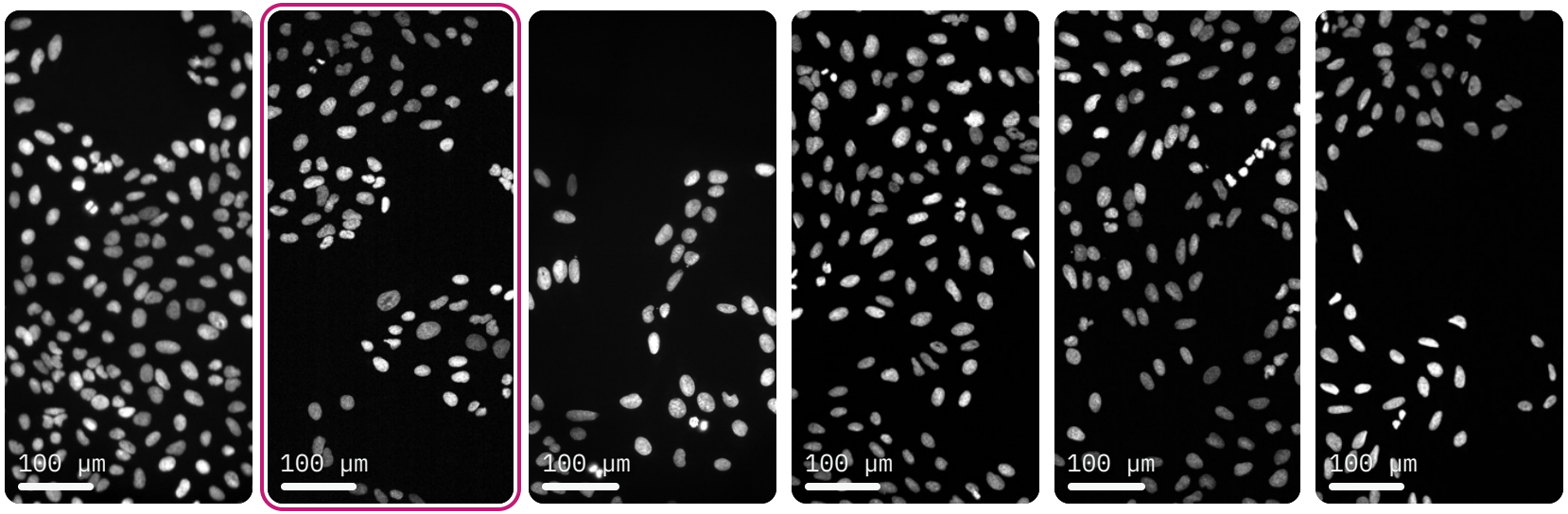
only control wells leave a lab, DMSO and known drugs, never a test compound, privacy A18, the lab’s share A24

Biology's ugliest dataset, built on purpose.

4 cell lines × 2 microscope types × known drugs
+ planted faults. Every correction method scored.

TODAY, SAME CELLS, SAME PROTOCOL, NO DRUG

real wells, JUMP [7]



LAB 1	LAB 2 (YOU)	LAB 3	LAB 5	LAB 6	LAB 10
Phenix, widefield	CV8000, confocal	Phenix, widefield	CV8000, confocal	CV8000, confocal	CV8000, confocal

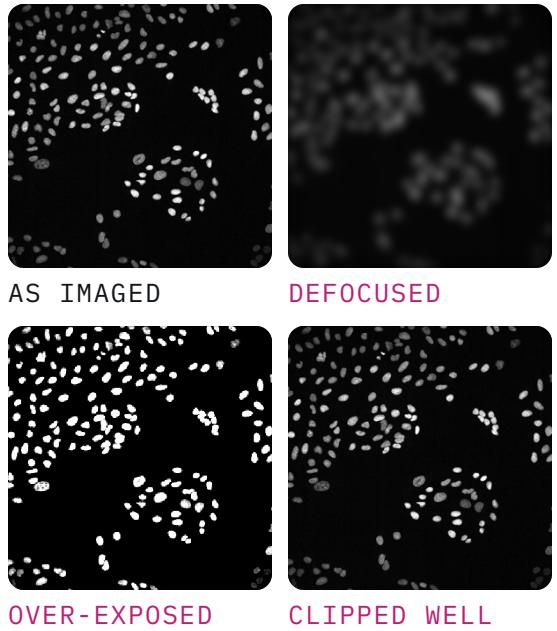
5 microscope models

8 plate layouts

13 labs [7]

PLANTED FAULTS

simulated



AS IMAGED

DEFOCUSED

OVER-EXPOSED

CLIPPED WELL

Can a model name the fault?

cell lines: U2OS, A549, HepG2, iPSC [17], microscope types: widefield + confocal

Ranked on controls, never on results.

Every lab scored on the other labs' scale, never its own.
Lab 2 (you): 8th of 11 labs.

LAB	BATCHES	SCORE	BEST RANK	CLEAN RUN	HOW UNUSUAL	PLATES DIFFER	BATCH TO BATCH	CELL COUNT OFF
lab 5	33	**70**	#1	33	3.35	0.27	0.81	0.31
lab 6	13	**60**	#16	13	4.44	0.41	1.23	0.12
lab 10	17	**54**	#10	17	3.84	0.62	1.21	0.23
lab 8	4	**52**	#54	4	4.10	1.08	0.86	0.09
lab 11	5	**47**	#23	2	3.79	0.87	1.57	0.17
lab 3	13	**44**	#18	6	3.27	0.80	2.01	0.29
lab 7	8	**42**	#69	8	6.38	0.58	1.17	0.33
lab 2	13	**39**	#48	10	5.68	0.57	1.99	0.14
lab 1	6	**32**	#78	6	8.68	1.10	2.03	0.09
lab 9	11	**25**	#97	11	9.97	0.50	1.76	0.45
lab 15	6	**23**	#68	6	5.53	0.87	1.69	0.44

SCORE

0 to 100, higher is better

all labs: how unusual 4.01, plates differ 0.51, batch to batch 1.17, cell count off 0.28
average rank on the four measures, among 129 batches
measures: lower is better, green beats all labs

BEST BATCH

82

lab 5, 28 Jun, 7 plates
lab 5 holds 9 of top 10

YOUR CHOICE TO PLAY

Named only if you opt in.
Private version for your own plates.

Lab medians, 1,871 plates, 11 labs, clean run: batches in a row never flagged (A11), JUMP [7]

Only control wells leave the lab.

No compound, no target, no result.
Nothing for legal to approve.

WHAT LEAVES THE BUILDING

control wells only: DMSO and known drugs



pink: 64 of 384 wells leave, JUMP [7]

images, or numbers from them

proposed: images deleted once measured

a plate map and the batch date

which wells, and when they were imaged

WHAT DATUM LINE DOES WITH IT

builds the map

one average point per plate.
Scale set by the other labs, never by yours.

the map holds DMSO wells only

82,319

for lab 2: DMSO wells of 10 other labs
0 treated

answers

where you sit, how far you moved, nearest lab, cell count, “normal” or not

ranks every batch

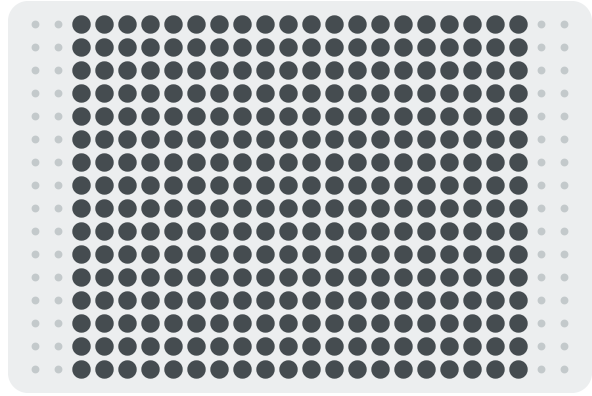
on its DMSO wells, never on results, A17

sends a correction, not a model of you

one per batch, fitted on your DMSO wells, A19

WHAT IT NEVER SEES

your treated wells



dark: 320 of 384 wells stay with you

your results

corrected in your building, by you

your name, unless you choose

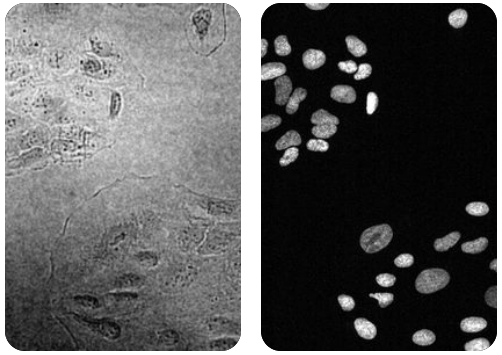
proposed: a number until you opt in.
Every lab here is a number.

In every analysis here, the map and the correction see DMSO wells only, github.com/kat-titter/datum-line/analysis, JUMP [7]

An image goes in; a position and a correction come out.

Results stay in the lab.
DMSO wells and the correction travel.

1, AN IMAGE



brightfield DNA stain

one DMSO well, JUMP [7]

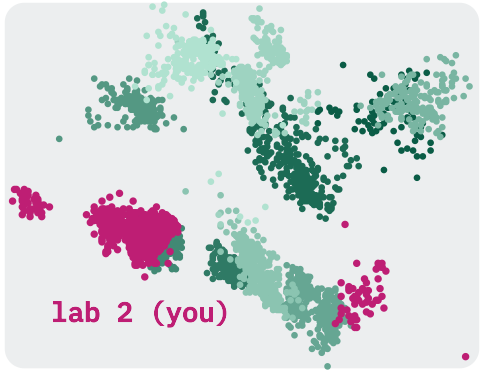
2, NUMBERS

3,773

numbers per well

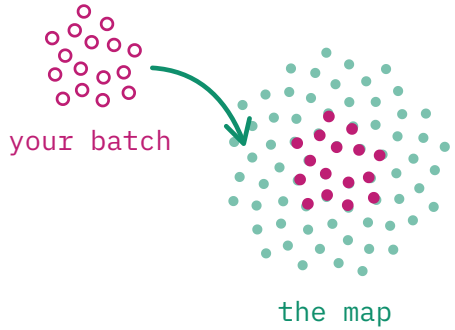
standard software today: size, shape, brightness, texture

3, THE MAP



each lab placed on the other labs' scale, never its own

4, THE CORRECTION



one per batch, fitted on your DMSO wells, sketch, A14

EVERY LAB LOOKS DIFFERENT

99.9%

a computer names the lab of each well, by guessing: 9%
93,228 DMSO wells, 11 labs, on plates it never saw

AI IMAGE MODEL NEXT, A10

brightfield, 8 labs 99.5%

DNA stain, 8 labs 96.3%

right lab, from one picture

Correction, per batch on 200 components: $x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$, every script: github.com/kat-titter/datum-line/analysis

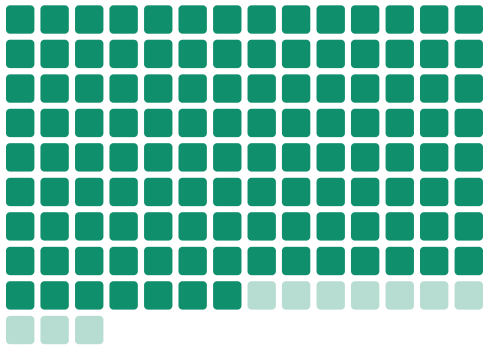
Every lab leaves a fingerprint, so a wrong label stands out.

One map of DMSO wells, no results stored.
Two checks work today, two are proposed.

PROVENANCE today

99.9%

of 93,228 DMSO wells get the right lab, on unseen plates. The fingerprint is cells and microscope together: image-quality numbers alone name the lab at 99.9%.



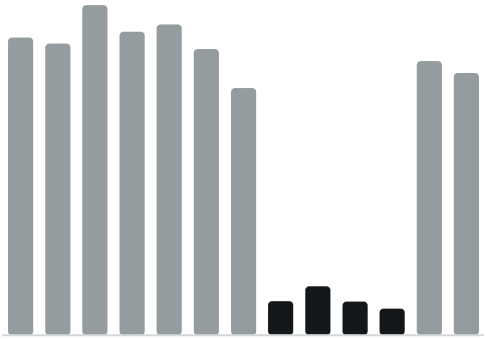
1 square = 1 batch
solid: every well right, 119

INTEGRITY today

4 batches

Label: known drug in these wells.

Cells: almost no effect, on 65 plates.



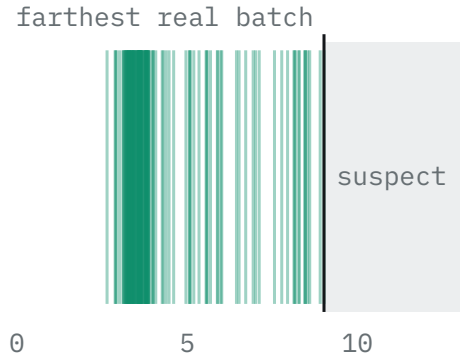
lab 3, drug effect per batch
dark: failed, A8

UNKNOWN ORIGIN proposed

9×

Farthest of 129 real batches from its nearest other lab.

Farther out: wrong cell line? Mixed culture? Untested.



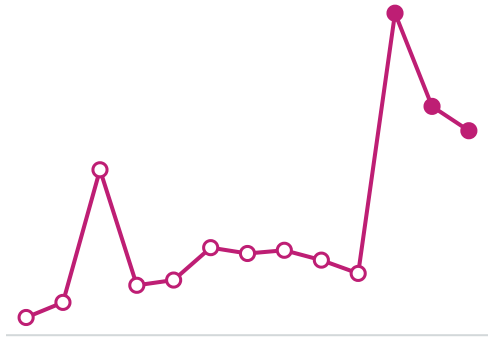
distance to nearest other lab
1 line = 1 real batch

EARLY WARNING proposed

30 Aug

2021: lab 2's DMSO wells moved; its cell count did not.

A move nobody explains: check reagents, air, water.



lab 2 (you), since batch 1
filled: flagged, A13

Integrity check uses each plate's known-drug wells (A8), × = times the usual spread of wells within one plate, JUMP [7]

proposed, every line an assumption

About $1.8M for eighteen months, mostly people.

$884k planned × 2 = $1.77M asked



people 71% | benchmark 17% | other 12%

leader, to run it	18 months	**$180k**
scientific lead (me)	18 months	**$180k**
ML engineer	from month 7	**$160k**
payroll overhead	20% of salaries	**$104k**
benchmark playground	~60 plates, planted faults, A16	**$120k**
partner-lab bench time	3 labs, controls they already image, reagent top-ups, shipping	**$30k**
compute and storage	images, models, hosting, API	**$45k**
legal, IP, insurance, accounting	incorporation to first contract	**$45k**
travel and partner visits	2 benchmark partners, 3 labs	**$20k**
planned	sum	**$884k**
+ buffer	most things take 2–3× the time and money planned, we plan for 2×	$884k
asked, 18 months		**$1.77M**

NO IN-HOUSE LAB

Partner labs or a contract lab.
Own lab: costs more, proves less.

WHAT THE MONEY MUST SHOW

1 benchmark playground, paid in this round **2** three partner labs on the map
3 model for any image **4** first paid certificate → seed

salaries below market, leader and scientific lead paid, nothing here committed

About $84M a year, if every likely buyer joined.

Count × price.
Three streams.

assumption	low	base	high
organisations testing drugs on a few cell lines	200	400	800
cell lines each one uses per quarter	2	4	8
price per cell line per quarter, $k	5	10	15
organisations testing drugs on hundreds to thousands of cell lines	10	25	50
site licence per year, $k	50	150	300
per-line checks and site licences, per year	**$8.5M**	**$67.8M**	**$399M**
companies buying cell data they did not make	50	150	300
purchased datasets each certifies per year	2	4	8
price per certificate, $k	10	25	50
certificates on purchased data, per year	**$1M**	**$15M**	**$120M**
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
benchmark membership, per year	**$200k**	**$1M**	**$4M**
total per year, if every buyer joined	**$9.7M**	**$83.8M**	**$523M**

BASE CASE, PER YEAR



↑ five-year target, a tenth, $8.4M

benchmark $1M ↑

WHY BUYERS PAY

One screen I ran on a line checked late:
3 runs a week × 17 weeks × $50k ≈ $2.6M to redo.
Certificate @ $25k: about 1% of that.
Subscription: insurance against it.

prices proposed on A15, academic labs join free, they are the map, not the revenue

Cell data sells, but across labs it is not one dataset.

Three public prices.
Same cells, four labs, different results.

AI TRAINING DATA

$1.5M a day

paid to experts for AI training data, by one vendor

~$85 an hour, $10B valuation [10]

CELL MAPS

$30M

paid by Roche for one cell dataset

second milestone, $213M to date [11]

CONSORTIA

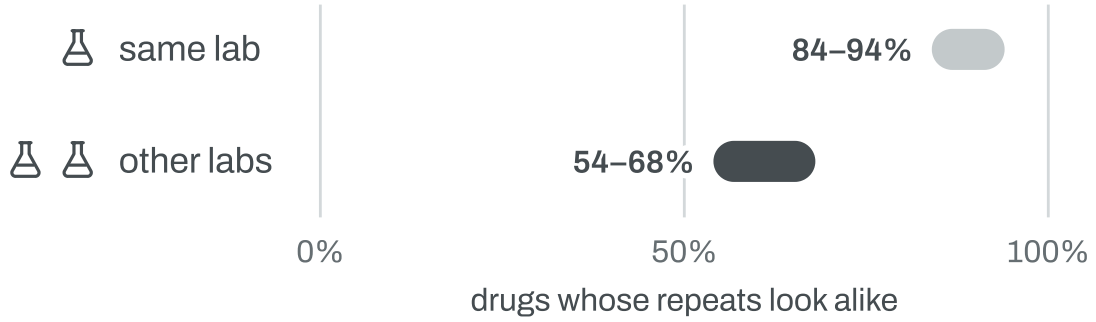
$1.8B

pledged for cell data to train AI
funders get it a year early

Meta, DeepMind, Isomorphic: $300M [12]

4 EUROPEAN LABS, ONE PROTOCOL

same cells, same reagents, shipped to every lab [13]



WHAT THE PAPER SAYS

“not directly comparable between imaging sites” [13]

ON ONE SHARED MAP, JUMP [7]

107 of 125

batches: labs agree more than on their own controls, A14

[7] JUMP, 2023, [10] TechCrunch, 2025, [11] Recursion, 2025, [12] eWeek, 2026, [13] Wolff et al., *iScience*, 2025, full references in the README

The lab keeps its data and a share; anyone can check it.

Subscribers get access, not ownership.
No crypto token.

CONTROL WELLS ONLY — today

no test compound, no target, no result
results never leave the lab

LAB KEEPS ITS DATA — proposed

sharing is opt-in
withdraw any time

LOW RISK BY DESIGN — today

cell-line images only
no human genetic data, no pathogen data

LABS LEAVE FINGERPRINTS — today

99.9% right lab out of 11
from control-well images A20

SIGNED AND TIMESTAMPED — proposed

public record, anyone can check
dates cannot be changed, blockchain optional

KNOWN SUBSCRIBERS — proposed

verified at sign-up, odd access flagged
access log open to members

lab shares
images of control cells
certificate attached

→

certificate checks
how “normal” the cells look
which lab, what changed

→

subscriber pays
to use certified data
from many labs

→

lab keeps a share
of every subscription
as long as its data is used

Every lab's controls, as one dataset each, with its certificate.

U2OS controls shared by 11 labs, replayed on public data. A certificate travels with every batch.

LAB	PLATES	BATCHES, IN RUN ORDER	CERTIFIED
lab 3	237	■■■■■□□□□□■■	8 of 13
lab 6	225	■■■■■■■■■■■■■	13 of 13
lab 2 (you)	219	■■■■■■■■■□□□	10 of 13
lab 10	205	■■■■■■■■■■■■■■■■■	17 of 17
lab 8	199	■■■■	4 of 4
lab 5	197	■■■■■■■■■■■■■■■■■■■■■■■■■	
lab 11	172	■■□■■	4 of 5
lab 15	146	■■■■■■	6 of 6
lab 7	121	■■■■■■■■	8 of 8
lab 9	99	■■■■■■■■■■■	11 of 11
lab 1	51	■■■■■■	6 of 6

batches certified = on the map and known drugs worked, 1,871 plates, JUMP [7]

WHAT A SQUARE IS

one batch of a lab's DMSO wells; filled: on the map and its known drugs worked; hollow: off the map, or a known drug failed (A8, A11)

WHERE IT CAME FROM

every well traced to its lab by its images: 99.9% named right of 11 labs (A20)

WHO CAN USE IT

subscribers, for access to certified data from many labs; the lab keeps its data and a share of every subscription (proposed, A15, A24)